ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION WINTER SEMESTER, 2018-2019
DURATION: 3 Hours FULL MARKS: 150

CSE 4305: Computer Organization and Architecture

Programmable calculators are not allowed. Do not write anything on the question paper.
There are **8 (eight)** questions. Answer any **6 (six)** of them.
Figures in the right margin indicate marks.

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1. a) Write down the basic differences between the followings with suitable examples: 6
   i. Computer organization and computer architecture
   ii. Computer structure and computer function

   b) On the IAS, appositely describe the process that the CPU must undertake stepwise to read a value from memory and to write a value to memory in terms of what is put into the MAR, MBR, address bus, data bus, and control bus with necessary figures. 10

   c) A pipelined processor has a clock rate of 2.5 GHz and executes a program with 1.5 million instructions. The pipeline has five stages, and instructions are issued at a rate of one per clock cycle. Ignore penalties due to branch instructions and out-of-sequence executions. 9
   i. What is the speedup of this processor for this program compared to a non-pipelined processor?
   ii. What is throughput (in MIPS) of the pipelined processor?

2. a) The instruction execution may involve several operations and depends on the nature of the instruction. Draw an instruction cycle state diagram with the interrupts which is required to comply in program execution cycle. Briefly describe all the states with a suitable example. 10

   b) What happens when a check bit rather than a data bit is in error? How many check bits are needed if the Hamming error correction code is used to detect single bit errors in a 1024-bit data word? Justify your answer. 7

   c) Consider a computer system that contains an I/O module controlling a simple key-board/printer teletype (an electromechanical device that can be used to send and receive typed messages through various communications channels). The following registers are contained in the processor and connected directly to the system bus: 8

   INPR: Input Register, 8 bits
   OUTR: Output Register, 8 bits
   FGI: Input Flag, 1 bit
   FGO: Output Flag, 1 bit
   IEN: Interrupt Enable, 1 bit

   Keystroke input from the teletype and printer output to the teletype are controlled by the I/O module. The teletype is able to encode an alphanumeric symbol to an 8-bit word and decode an 8-bit word into an alphanumeric symbol. Now answer the following questions:
   i. Describe how the processor, using the first four registers listed in this problem, can achieve I/O with the teletype.
   ii. Describe how the function can be performed more efficiently by also employing IEN.

3. a) Define logical cache and physical cache. Comparatively distinguish them with necessary figures. 8

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b) Suppose that the processor has access to two levels of memory. Level 1 contains 1000 words and has an access time of 0.01 μs; level 2 contains 100,000 words and has an access time of 0.1 μs. Assume that if a word to be accessed is in level 1, then the processor accesses it directly. If it is in level 2, then the word is first transferred to level 1 and then accessed by the processor. For simplicity, we ignore the time required for the processor to determine whether the word is in level 1 or level 2. If 95% of the memory accesses are found in level 1, what is the average time to access a word? 5

c) Consider a machine with a byte addressable main memory of 2^{16} bytes and block size of 8 bytes. Assume that a direct mapped cache consisting of 32 lines is used with this machine. 12

i. How is a 16-bit memory address divided into tag, line number, and byte number?

ii. Into what line would bytes with each of the following addresses be stored?

0001 0001 0001 1011
1100 0011 0011 0100
1101 0000 0001 1101
1010 1010 1010 1010

iii. Suppose the byte with address 0001 1010 0001 1010 is stored in the cache. What are the addresses of the other bytes stored along with it?

iv. How many total bytes of memory can be stored in the cache?

v. Why the tag is also stored in the cache?

4. a) Magnetic disk is the foundation of external memory on virtually all computer systems. Briefly describe how data are recorded on and later retrieved from the disk with necessary figures. 8

b) Write short notes on following terms: 5

i. Data striping
ii. RAID 10
iii. CD-R
iv. MZR
v. Pit

c) Consider a single-platter disk with the following parameters: rotation speed: 7200 rpm; number of tracks on one side of platter: 30,000; number of sectors per track: 600; seek time: one ms for every hundred tracks traversed. Let the disk receive a request to access a random sector on a random track and assume the disk head starts at track 0. 12

i. What is the average seek time?
ii. What is the average rotational latency?
iii. What is the transfer time for a sector?
iv. What is the total average time to satisfy a request?

5. a) The processor needs to save information stored in several registers in system stack to resume the current program at the point of interrupt. When interrupt processing is complete, the saved register values are retrieved from the stack and restored to the registers. Now draw several figures to explain the above mentioned scenario with a portion of main memory and different registers of processor. 7

b) To implement interrupt driven I/O, respective design issues are raised. Mention those design issues and also briefly describe different techniques to solve them. 10

c) What is I/O channel? How many different types of I/O channels are there? Describe them with respective figures. 8

6. a) Many instruction sets contain the instruction NOOP, meaning no operation, which has no effect on the processor state other than incrementing the program counter. Suggest some uses of this instruction. 5

b) Compare one-, two-, and three- address machines by writing programs to compute the following expression for each of the three machines. 12

$$X = (A + B * C) / (D - E * F)$$

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c) For the following data structure (figure 1), draw the big-endian and little-endian layouts, using the values given as comment: 8

```
struct {
   short i;      //0x1112
   short j;      //0x1314
   short k;      //0x1516
   short l;      //0x1718
} s3;
```

Figure 1: A data structure

7. a) An instruction format defines the layout of the bits of an instruction. The design of an instruction format is a complex art, and an amazing variety of designs have been implemented. There are several design issues in this context. Briefly describe those issues mentioning different trade-offs. 8

b) A sharp contrast to the instruction set of the PDP-8 is that of the PDP-10 which was designed to be a large-scale time-shared system, with an emphasis on making the system easy to program. Briefly describe different design principles employed in designing the instruction set of PDP-10. 5

c) Given the following memory values and a one- address machine with an accumulator, what values do the following instructions load into the accumulator? 12

- Word 20 contains 40.
- Word 30 contains 50.
- Word 40 contains 60.
- Word 50 contains 70.

i. LOAD IMMEDIATE 20
ii. LOAD DIRECT 20
iii. LOAD INDIRECT 20
iv. LOAD IMMEDIATE 30
v. LOAD DIRECT 30
vi. LOAD INDIRECT 30

8. a) With the earliest computers, the programmer interacted directly with the computer hardware without OS. These early systems presented two main problems. Briefly describe those problems with necessary examples. 5

b) Comparatively describe partitioning, paging and segmentation in memory management. Among them, segmentation has a number of advantages to the programmer over a non-segmented address space. Briefly remarks on those advantages. 10

c) List and briefly explain various ways in which an instruction pipeline can deal with conditional branch instructions. 10

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